RUSHI ASHOK SAVANI

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WORK EXPERIENCE

AI Researcher | Syracuse University, New York **Sep 2026 – Present**

- Analyzed research datasets using **Python, SQL** and statistical methods to identify trends, investigate relationships between variables and develop findings for academic research.
- Conducted **hypothesis tests** to assess variable significance, informing feature selection, model selection and validation approaches for machine learning experiments.

AI/ML Engineer | ProHealth Group, Atlanta, Georgia **Jun 2025 – May 2026**

- Collaborated with leadership and clinicians to identify **AI automation opportunities** through operational analysis and analytical dashboards, translating workflow inefficiencies into actionable solutions.
- Architected and deployed a **Text-to-SQL agent** using **LangGraph, MCP** and hybrid retrieval, enabling natural-language queries across patient and operational records, reducing information retrieval time by **90%**.
- Automated **3+ operational and financial investigation workflows** using Python, LLMs, and **LangChain**, delivering auditable root-cause reports that accelerated anomaly detection and saved **$90K** quarterly.
- Implemented **LLM evaluation and security frameworks** using LangSmith, MLflow and Postman to investigate the AI response against **100+ test cases** for accuracy, hallucinations, access control, and prevention of unauthorized database operations.
- Developed a patient **rehospitalization predictive model** on AWS **SageMaker** using Python, Scikit-learn and clinical feature engineering, distinguishing preventable high-risk cases to prioritize targeted intervention with **83%** recall.
- Validated model impact through causal inference and randomized **A/B testing** against standard care plans, cutting readmission cases by **10% (~$15K saving monthly)**.
- Deployed scalable AI services on AWS using **Kubernetes, docker and CI/CD**, reducing inference costs by **17%** through **Redis semantic caching**, incremental data processing and optimized LLM execution.

Research Assistant | Syracuse University, New York **Jan 2025 – May 2025**

- Analyzed **7,000+ employee survey responses** on enterprise AI adoption, automating data preprocessing to resolve missing values, standardize multilingual data, and prepare datasets for sentiment analysis.
- Fine-tuned **RoBERTa using PyTorch and Hugging Face Transformers** for study-specific sentiment classification, improving accuracy by **30%** while nearly eliminating manual labeling.
- Identified employee AI-adoption patterns using **Sentence-BERT embeddings** and **HDBSCAN clustering**, presenting behavioral segments and sentiment trends through **Tableau dashboards**.

Data Scientist | IBM, India, Mumbai **Jan 2022 - Jun 2024**

- Consulted **3 enterprise clients** on business challenges and ML opportunities, presenting **5+ solution proposals** decks to leadership and aligning recommendations with operational requirements.
- Delivered **risk-scoring and demand-forecasting models** using XGBoost, Random Forest, and Logistic Regression on Azure ML, improving baseline precision by **18%** and contributing to **$1.2M in operational savings**.
- Established end-to-end **ETL pipelines** using Azure Data Factory and SQL, integrating multiple data sources into staging and analytical marts to process **1M+ daily transactions** with **99.9% availability**.
- Assessed ML model performance through **real-world testing, statistical validation, and Bayesian optimization** using Optuna, addressing data leakage and accelerating model selection by **20%**.
- Communicated findings through **Tableau dashboards and executive presentations**, translating model outcomes into recommendations for non-technical stakeholders while improving internal PostgreSQL query performance by **40%**.

TECHNICAL SKILLS

Languages & Dev Tools: Python, SQL, TypeScript, Node.js, Claude Code, Cursor, Codex
Frameworks: PyTorch, TensorFlow, Hugging Face Transformer, FastAPI, LangChain/LangGraph, Google ADK, LlamaIndex, CUDA
ML & Deep Learning: Scikit-learn, Pandas, NumPy, XGBoost, Gradient Boosting, Regression, Classification, Tree models, SVM, Hyperparameter Tuning, Classification Metrics, Natural Language Processing (NLP), Sentiment Analysis
Generative AI: RAG, MCP, LLMs, Prompt Engineering, LiteLLM Inference, LangSmith, vLLM
MLOps & Cloud: AWS, Microsoft Azure, Docker, Kubernetes, HPA, GitHub Actions CI/CD, Redis, CloudWatch, Token-economics
Databases: SQL Server, Snowflake, Vector Store (pgvector), AWS Redshift
BI & Analytics: Power BI, Tableau, Streamlit, ETL/ELT, Data Modeling, Statistical Analysis, Hypothesis Testing, A/B Testing

EDUCATION

Syracuse University, New York - Masters | Information Systems, Data Science **Aug 2024 - May 2026**
Thadomal Shahani Engineering College, Mumbai, India - Bachelors | Information Technology **Jun 2020 - Jun 2024**

PROJECTS

Distributed LLM Inference Platform | Python, vLLM, PyTorch, GCP, Kubernetes

- Optimized a PyTorch CUDA engine on NVIDIA L4 (naive, KV cache, continuous batching, vLLM) at 214 vs 34 tok/s at concurrency 16, cut O(n^2) decode to O(n) (~5x), FP16/INT8/INT4 (980 to 489 MiB), GKE HPA 1 to 4 pods. | GitHub

PrivateRAG | Python, Qdrant, Llama 3.1, TypeScript, LlamaIndex

- Designed a fully local RAG system for private-document Q&A with Qdrant vector indexing, metadata filtering, and Llama 3.1 inference, featuring source attribution and chunk-level traceability. | GitHub

AI Email Response System | Python, RAG, MCP, LLM-as-a-Judge

- Implemented an automated RAG-based email response workflow with policy checks, MCP-based routing, LLM-as-a-Judge validation, and human handoff for high-risk cases, reducing manual investigation by 70%. | GitHub